

Task 2: Continuous Variables

COURSE NAME

CSDS 413 Introduction to Data Analysis

Authors:

Jacob Anderson

November 10, 2025

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Context

In this exercise, our aim is to quantify the associations between continuous variables and assess the statistical significance of these associations. For this purpose, we will use the three datasets that are provided with the assignment:

- The file "p2a.csv" contains a matrix of size 2400 x 2 which has 2400 samples and 2 variables.
- The file "p2b.csv" contains a matrix of size 110 x 2 which has 110 samples and 2 variables.
- The file "p2c.csv" contains a matrix of size 2100 x 2 which has 2100 samples and 2 variables.

1 Part (a)

For the two variables provided in "p2a.csv", assess the association between them by computing Pearson correlation r_a and computing a p-value p_a for the null hypothesis of no association. Select a significance level α and reject the null-hypothesis if the p-value is less than α . Explain, in complete sentences, your findings: Is there a statistically significant association (at α level) between the provided variables? What is the magnitude and the direction of the association?

2 Part (b)

Repeat part (a) for the variable pair provided in "p2b.csv" and compute Pearson correlation r_b and p-value p_b . Compare the Pearson correlations r_a and r_b as well as the p-values p_a and p_b . Explain your findings: Which variable pair (in part *a* or *b*) has a stronger association according to the comparison of the correlations? Which variable pair has a stronger association according to the comparison of the p-values? Do the comparisons according to correlation coefficients and p-values agree on which variable pair indicate the same stronger association? If not, why is there such a discrepancy? Next, draw scatter plots (variable 1 vs. variable 2) to visualize the data for both part (a) and part (b). Which variable pair (in part *a* or *b*) has a stronger association do you think according to the scatter plots? Does your conclusion agree with the comparisons of the p-values and correlation coefficients? If not, explain why.

3 Part (c)

Repeat part (b). But, this time compare the associations in the datasets "p2a.csv" and "p2c.csv". Make sure to answer the questions in part (b), this time for comparing the variable pairs in parts (a) and (c).